

CBSE Class 10 Mathematics — Statistics — Solutions

1. 1. The mean of the first 5 natural numbers is:

Answer: (B)

Solution:

1. List the first 5 natural numbers: 1, 2, 3, 4, 5.
2. Find the sum: $1 + 2 + 3 + 4 + 5 = 15$.
3. Divide by the count (5) to find the mean: $15 / 5 = 3$.

Derivation:

$$\text{Mean} = \frac{\sum x_i}{n} = \frac{1 + 2 + 3 + 4 + 5}{5} = \frac{15}{5} = 3$$

Statistics — Mean of ungrouped data

2. 2. The mode of the data set 2, 3, 4, 3, 5, 3, 6, 3 is:

Answer: (C)

Solution:

1. The mode is the observation that occurs most frequently.
2. Count the frequencies: 2 appears once, 3 appears four times, 4 appears once, 5 appears once, 6 appears once.
3. Since 3 has the highest frequency, it is the mode.

Derivation:

Frequency of 3 = 4, which is the maximum. Therefore, Mode = 3

Statistics — Mode of ungrouped data

3. 3. If the mean of 5 observations $x, x + 2, x + 4, x + 6, x + 8$ is 11, then the value of x is:

Answer: (A)

Solution:

1. Sum the observations: $x + (x + 2) + (x + 4) + (x + 6) + (x + 8) = 5x + 20$.
2. Set up the mean formula: $(5x + 20)/5 = 11$.
3. Solve for x : $5x + 20 = 55 \implies 5x = 35 \implies x = 7$.

Derivation:

$$\begin{aligned}\frac{5x + 20}{5} &= 11 \\ \implies x + 4 &= 11 \\ \implies x &= 7\end{aligned}$$

Statistics — Mean of ungrouped data

4. **4.** The median of the data 7, 3, 5, 9, 1 is:

Answer: (B)

Solution:

1. Arrange data in ascending order: 1, 3, 5, 7, 9.
2. Identify the middle term for $n = 5$ (odd), which is the $(n + 1)/2$ term.
3. The 3rd term is 5.

Derivation:

$$\text{Sorted: } 1, 3, 5, 7, 9. \text{ Median} = \left(\frac{5 + 1}{2} \right)^{\text{th}} \text{ term} = 3^{\text{rd}} \text{ term} = 5$$

Statistics — Median of ungrouped data

5. **5.** For a distribution, if mean = 20 and median = 21, then the mode is:

Answer: (D)

Solution:

1. Use the empirical relationship: $\text{Mode} = 3(\text{Median}) - 2(\text{Mean})$.
2. Substitute the values: $\text{Mode} = 3(21) - 2(20)$.
3. Calculate: $63 - 40 = 23$.

Derivation:

$$\text{Mode} = 3(21) - 2(20) = 63 - 40 = 23$$

Statistics — Empirical relationship between Mean, Median, and Mode

6. **6.** Assertion (A): The mean of the first five natural numbers is 3. Reason (R): The mean of a set of observations is calculated by dividing the sum of observations by the total number of observations.

Answer: (A)

Solution:

1. $\text{Sum of first five natural numbers} = 1 + 2 + 3 + 4 + 5 = 15$
2. $\text{Mean} = \text{Sum} / \text{Count} = 15 / 5 = 3$
3. The formula for mean is standard definition.

Derivation:

$$\text{Mean} = \frac{\sum x_i}{n} = \frac{1 + 2 + 3 + 4 + 5}{5} = \frac{15}{5} = 3$$

Statistics — Mean of ungrouped data

7. **7.** Assertion (A): The mode of the data 2, 3, 4, 3, 2, 3 is 3. Reason (R): Mode is the value that appears most frequently in a data set.

Answer: (A)

Solution:

1. Count the frequency of each number: 2 appears twice, 3 appears three times, 4 appears once.
2. The number with the highest frequency is 3.
3. Definition of mode is correctly stated in R.

Derivation:

Frequency of 2 = 2, Frequency of 3 = 3, Frequency of 4 = 1. $\therefore$ Mode = 3

Statistics — Mode of ungrouped data

8. **8.** Assertion (A): Median of the data 5, 1, 3, 2, 4 is 3. Reason (R): Median is the middle value when data is arranged in ascending or descending order.

Answer: (A)

Solution:

1. Arrange in ascending order: 1, 2, 3, 4, 5.
2. The middle value is the 3rd term, which is 3.
3. Definition of median is correctly stated in R.

Derivation:

Sorted data: 1, 2, 3, 4, 5. Middle term is 3.

Statistics — Median of ungrouped data

9. **9.** Assertion (A): The range of the data 10, 15, 20, 25, 30 is 20. Reason (R): Range is defined as the difference between the maximum and minimum values of the data.

Answer: (A)

Solution:

1. $Maxvalue = 30, Minvalue = 10$.
2. $Range = Max - Min = 30 - 10 = 20$.
3. Definition of range is correctly stated in R.

Derivation:

Range = 30 - 10 = 20

Statistics — Range

10. **10.** Assertion (A): The sum of deviations of the observations x_i from their mean $\bar{x}$ is 0. Reason (R): $\sum(x_i - \bar{x}) = 0$ is a fundamental property of the arithmetic mean.

Answer: (A)

Solution:

1. Expand the summation: $\sum x_i - \sum \bar{x}$.
2. Since $\sum \bar{x} = n \times \bar{x}$ and $\bar{x} = \frac{\sum x_i}{n}$, the expression becomes $\sum x_i - n(\frac{\sum x_i}{n}) = 0$.
3. The reason is the correct mathematical justification.

Derivation:

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum x_i - \sum \bar{x} = \sum x_i - n\bar{x} = \sum x_i - n\left(\frac{\sum x_i}{n}\right) = 0$$

Statistics — Properties of Mean

11. 11. Find the mean of the first n natural numbers.

Answer:

$$\text{Mean} = \frac{n+1}{2}$$

Solution:

1. Use the formula for the sum of first n natural numbers: $S_n = \frac{n(n+1)}{2}$.
2. Divide by the total count n to find the mean.

Derivation:

$$\text{Mean} = \frac{\sum x_i}{n} = \frac{n(n+1)}{2n} = \frac{n+1}{2}$$

Statistics — Mean of grouped data

12. 12. If the mean of the following distribution is 6, find the value of p : $x : 2, 4, 6, 10, p+5$; $f : 3, 2, 3, 1, 2$.

Answer:

$$p = 3$$

Solution:

1. Calculate $\sum f_i x_i = (2 \times 3) + (4 \times 2) + (6 \times 3) + (10 \times 1) + (p+5) \times 2$.
2. Set $\frac{\sum f_i x_i}{\sum f_i} = 6$ and solve for p .

Derivation:

$$\begin{aligned} 6 + 8 + 18 + 10 + 2p + 10 &= 6(3 + 2 + 3 + 1 + 2) \\ &\Rightarrow 52 + 2p = 6(11) \\ &\Rightarrow 2p = 66 - 52 = 14 \\ &\Rightarrow p = 7 \text{ (Correction: Calculation check } 52 + 2p = 66) \\ &\Rightarrow 2p = 14 \\ &\Rightarrow p = 7) \end{aligned}$$

Statistics — Mean

13. 13. The median of a distribution is 15 and the mean is 12. Find the mode using the empirical relationship.

Answer:

$$\text{Mode} = 21$$

Solution:

1. Use the empirical formula: $\text{Mode} = 3(\text{Median}) - 2(\text{Mean})$.

2. Substitute the given values and solve.

Derivation:

$$Mode = 3(15) - 2(12) = 45 - 24 = 21$$

Statistics — Relationship between Mean, Median and Mode

14. **14.** Find the class mark of the class interval 10 – 25 and 35 – 50.

Answer:

$$17.5 \text{ and } 42.5$$

Solution:

1. Use the formula: $ClassMark = \frac{UpperLimit + LowerLimit}{2}$.
2. Apply for both intervals.

Derivation:

$$\frac{10 + 25}{2} = 17.5; \frac{35 + 50}{2} = 42.5$$

Statistics — Grouped Data

15. **15.** If $\sum f_i = 20$ and $\sum f_i x_i = 2p + 120$, find p if the mean is 7.

Answer:

$$p = 10$$

Solution:

1. Use Mean formula $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$.
2. Substitute $7 = \frac{2p+120}{20}$ and solve for p .

Derivation:

$$\begin{aligned} 140 &= 2p + 120 \\ \Rightarrow 2p &= 20 \\ \Rightarrow p &= 10 \end{aligned}$$

Statistics — Mean

16. **16.** The mean of the following frequency distribution is 50. Find the value of p if the sum of frequencies is 100. — Class — 0-20 — 20-40 — 40-60 — 60-80 — 80-100 —
————— Frequency — 17 — p — 32 — p — 19 —

Answer:

$$p = 16$$

Solution:

1. Calculate the class marks (x_i) for each interval.
2. Use the formula Mean = $\frac{\sum f_i x_i}{\sum f_i}$.
3. Substitute the given mean and sum of frequencies to form a linear equation in p and solve.

Derivation:

$$\Rightarrow p = 16. \text{Verification: } \sum f_i x_i = (17 \times 10) + (16 \times 30) + (32 \times 50) + (16 \times 70) + (19 \times 90) = 1$$

Statistics — Mean of Grouped Data

17. 17. The following table gives the daily income of 50 workers of a factory. Find the mean and daily income of

120	120 – 140	140 – 160	160 – 180	180 – 200		– – –		– – –		– – –		– – –		– – –
– – –														
Number of workers 12 14 8 6 10														

Answer:

Rs. 145

Solution:

1. Choose an assumed mean $a = 150$ (midpoint of the middle class).
2. Calculate deviations $d_i = x_i - 150$ and find $\sum f_i d_i$.
3. Apply the formula $\text{Mean} = a + \frac{\sum f_i d_i}{\sum f_i}$.

Derivation:

$$x_i : 110, 130, 150, 170, 190. d_i : -40, -20, 0, 20, 40. f_i d_i : -480, -280, 0, 120, 400. \sum f_i d_i = -240. \text{Mean}$$

Statistics — Mean of Grouped Data

18. 18. Find the median of the following data: — Class interval — 0-10 — 10-20 —
 20-30 — 30-40 — 40-50 — ————— — Frequency — 5 — 15 —
 20 — 7 — 3 —

Answer:

25

Solution:

1. Calculate the cumulative frequency (cf) for each class.
2. Identify the median class where $N/2 = 50/2 = 25$ lies.
3. Apply the median formula $l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$.

Derivation:

$$N = 50, N/2 = 25. \text{Median class is } 20-30. l = 20, cf = 20, f = 20, h = 10. \text{Median} = 20 + \frac{25 - 20}{20} \times 10$$

Statistics — Median of Grouped Data

- 19. 19.** Find the modal class and the mode for the following distribution: — Marks —
 0-10 — 10-20 — 20-30 — 30-40 — 40-50 — ————— — Number
 of students — 4 — 10 — 16 — 12 — 8 —

Answer:

$$\text{Modal class} : 20 - 30, \text{Mode} : 25.71$$

Solution:

1. Identify the class with the highest frequency as the modal class.
2. Use the formula $\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$.
3. Substitute values and calculate.

Derivation:

$$\text{Modal class} = 20-30. l = 20, f_1 = 16, f_0 = 10, f_2 = 12, h = 10. \text{Mode} = 20 + \frac{16 - 10}{32 - 10 - 12} \times 10 = 20 +$$

Statistics — Mode of Grouped Data

- 20. 20.** If the median of a distribution is 28 and the mean is 27, find the mode using the empirical relationship.

Answer:

$$30$$

Solution:

1. State the empirical relationship between mean, median, and mode.
2. Substitute the given values into the formula.
3. Solve for the mode.

Derivation:

$$\text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean}. \text{Mode} = 3(28) - 2(27) = 84 - 54 = 30.$$

Statistics — Relationship between Mean, Median, and Mode

- 21. 21.** The following distribution shows the daily pocket allowance of 50 children of a locality. If the mean pocket allowance is {Rs. }18, find the missing frequency f for the class interval 19 – 21 : *Daily allowance (in Rs.)* : 11 – 13, 13 – 15, 15 – 17, 17 – 19, 19 – 21, 21 – 23, 23 – 25 *Number of children* : 7, 6, 9, 13, f , 5, 4

Answer:

$$f = 6$$

Solution:

1. Determine the class marks (x_i) for each interval.
2. Assume a mean (A) to use the Assumed Mean Method.
3. Calculate the deviations $d_i = x_i - A$.
4. Calculate $f_i d_i$ and find $\sum f_i$ and $\sum f_i d_i$.
5. Apply the formula $\text{Mean} = A + \frac{\sum f_i d_i}{\sum f_i}$ and solve for f .

Derivation:

$$\begin{aligned}\text{Let } A = 18. \sum f_i = 44 + f. \sum f_i d_i = 2f - 20.18 &= 18 + \frac{2f - 20}{44 + f} \\ \implies 0 &= 2f - 20 \\ \implies f &= 10\end{aligned}$$

Statistics — Mean of Grouped Data

- 22. 22.** The annual profits earned by 30 shops of a shopping complex in a locality are given below: Profit (in lakhs): 0-10, 10-20, 20-30, 30-40, 40-50, 50-60 Number of shops: 5, 8, 6, 4, 3, 4 Draw both 'less than' and 'more than' ogives for the given data on the same graph and find the median from the intersection point of the ogives.

Answer:

$$\text{Median} = 24$$

Solution:

1. Create a cumulative frequency table for 'less than' type.
2. Create a cumulative frequency table for 'more than' type.
3. Plot points (10, 5), (20, 13), ... for less than ogive.
4. Plot points (0, 30), (10, 25), ... for more than ogive.
5. Identify the x-coordinate of the intersection point.

Derivation:

Intersection of $F_<$ and $F_>$ at $x = 24$ gives the median.

Statistics — Graphical Representation of Cumulative Frequency

- 23. 23.** If the median of the following frequency distribution is 28.5, find the values of x and y if the total frequency is 60: Class Interval: 0-10, 10-20, 20-30, 30-40, 40-50, 50-60 Frequency: 5, x , 20, 15, y , 5

Answer:

$$x = 8, y = 7$$

Solution:

1. Set up the cumulative frequency table in terms of x and y .
2. Use the total frequency equation $45 + x + y = 60$.
3. Identify the median class based on the given median 28.5.
4. Use the median formula: $M = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$.
5. Substitute values and solve the system of two linear equations.

Derivation:

$$\begin{aligned}45 + x + y &= 60 \\ \Rightarrow x + y = 15; 28.5 &= 20 + \frac{30 - (5 + x)}{20} \times 10 \\ \Rightarrow 8.5 &= \frac{25 - x}{2} \\ \Rightarrow 17 &= 25 - x \\ \Rightarrow x = 8, y &= 7\end{aligned}$$

Statistics — Median of Grouped Data

- 24. 24.** The frequency distribution of the heights of 50 students is given below. Find the modal height of the students: Height (cm): 140-145, 145-150, 150-155, 155-160, 160-165 Number of students: 8, 12, 15, 10, 5

Answer:

$$151.67cm$$

Solution:

1. Identify the modal class (class with highest frequency).
2. Determine l, f_1, f_0, f_2 , and h .
3. Apply the mode formula: $\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$.
4. Perform the arithmetic calculation.
5. State the final result with units.

Derivation:

$$l = 150, f_1 = 15, f_0 = 12, f_2 = 10, h = 5. \text{Mode} = 150 + \frac{15 - 12}{30 - 12 - 10} \times 5 = 150 + \frac{3}{8} \times 5 = 150 + 1.875 =$$

Statistics — Mode of Grouped Data